

Enhanced nanoparticle uptake into virus infected cells: Could nanoparticles be useful in antiviral therapy?

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Roadmap:

Summary of Paper:

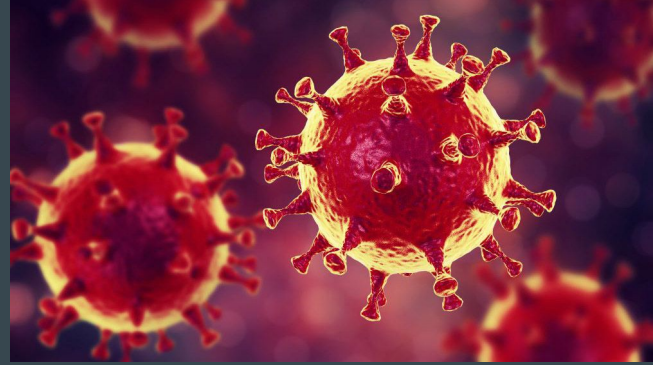
- Background
- Hypothesis
- Methodology
- Results/Conclusion

Proposed Research:

- Rationale
- Aims
- Challenges
- Methods

Background:

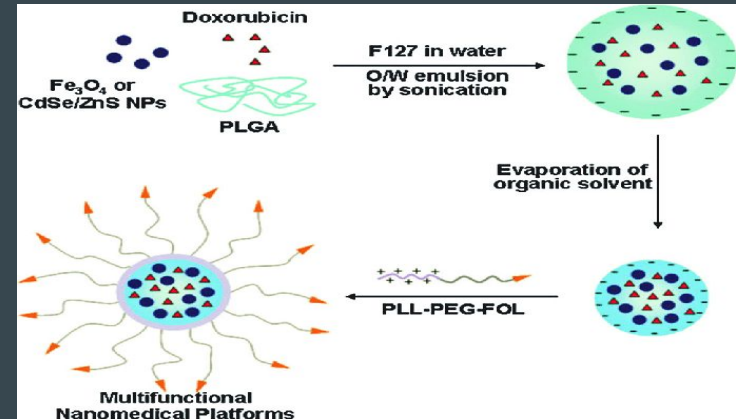
Virus:



- An infective agent that typically consists of a nucleic acid molecule in a protein coat, is too small to be seen by light microscopy and is able to multiply only within the living cells of a host
- Usually does not have cures due to accumulation at off target organs and the body can develop virus resistance due to virus mutations.
 - Many antiviral drugs available today but they come with lots of side effects
 - Ribavirin: causes haemolytic anaemia
 - Lamivudine, abacavir: can rise to peripheral neuropathy, pancreatitis, gout and etc

Purpose

- Developing a delivery system that can selectively target drug into affect organs and avoid off target accumulation
 - Example: delivering nanoparticles to target only the liver can improve hepatic virus infections such as Hep A, Hep B, and Hep C
- Hypothesis:
 - Better therapeutic efficacy with fewer side effects can be achieved



Pros

- Nano delivery is known to modify the physicochemical properties
 - Example:
 - Solubility and pharmacokinetic parameters of encapsulated materials via controlled drug release.
 - such as absorption, distribution, metabolism, and elimination
- Nano delivery could result in lower dose of antiviral agents and reduced side effect profile.
- Nanocrystals
 - A nanosized delivery system with 100% drug and no carrier
 - Showed results in reduced cytotoxicity compared to the free parent drug and improved cellular uptake into monocyte-derived macrophage

Cons

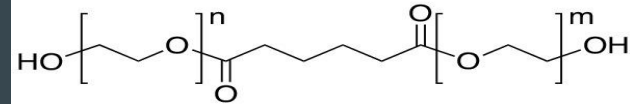
- The effect of virus infections on the uptake of NP into host cells has not been studied in detail.
- A downregulation of endocytosis by virus would make nanoparticle delivery of appropriate antiviral agents less effective whereas upregulation of endocytosis in infected cells can reduce side effects and facilitate targeted uptake of drug in infected cells.

Mechanism:

- They are using an established in-vitro model of hepatitis C virus (HCV) replication to interrogate the capacity of virus infection to modulate uptake of NP
- Will use Rhodamine B Isothiocyanate (RBITC), a fluorescently labelled polymer NP, prepared with with poly-glycol adipate (PGA) used in combination with Huh7.5 cells
 - Was transfected with HVC (J6/JFH1 chimera) to track how the virus infection could modulate the endocytic uptake of NP
- Huh7.5 cells are highly permissive to HCV and J6/JFH1 chimera (HCV RNA) is the most efficient replicating chimera in Huh7.5 cells

PGA

- A functionalized linear polyester
- enzymatically synthesized
 - Taking equal amounts (250mmol) of glycerol and DVA
 - Dissolved in dry tetrahydrofuran (THF, 30ml) with novozyme 435 (1.25g)
 - Was constantly stirred at 200rpm and kept at constant temperature of 50C for 24 hours
 - An enzyme filtration and evaporation of THF to obtain a yellowish-jelly-like polymer.



Fluorescently Labelled Nanoparticles

- RBITC was added to the aqueous phase
- The polymer (20mg) dissolved in acetone (2ml) added dropwise into the aqueous phase while stirring. Then leave the organic solvent was evaporated overnight.
- RBITC PGA NP was purified by loading the sample onto a Sepharose CL-4B gel column and the eluted by water .
- The peaks of dye labelled NP and free dye were detected using pharmacia chromatographic UV detector (206nm filter)
- The dye loading and encapsulation were then determined and acetone was used to extract a weight amount of freeze dried RBITC PGA NP

Stabilizing the NP in physiological buffer

- Used surfactants and human plasma to stabilize the RBITC PGA NP to physiological salt concentrations
 - Polysorbate surfactants were added to give final concentrations of 0.01% to 0.1% v/v to the pre-prepared NP
 - Freshly isolated human plasma was incubated with NP for 5 min
- RBITC PGA NP was diluted with PBS and the final volume for all the previous samples were adjusted to 1ml
- In order to get particle stability measurement, NP was diluted w/ HEPES buffer to 1 ml and used as a control for NP size

NP uptake by HCV transfected/non-transfected Huh7.5 cells

- Huh7.5 cells were electroporated in cytomix buffer(400microL)
- Cells were then incubated in Phenol red free- DMEM and then they were added to a 24-well cluster containing glass cover slips then incubated for another 24h
- Phenol red free-DMEM (0.8ml) and NP stabilised with FBS were then added to these cells then incubated again @ 37C
- Cells were then washed with PBS fixed using paraformaldehyde and then permeabilized with 0.5% triton x-100 for 5 min

- The virus core protein staining was carried by addition of mAb C7-50
- incubated and washed with PBS
- then incubated again with alexa-88 anti mouse IgG at room temperature in 5% FBS in PBS (60 min) and then washed 3 times with PBS
- A set of virus transfected/non-transfected Huh7.5 cells that were not treated were NP were prepared and processed under similar conditions and imaged by fluorescence microscopy to investigate the efficiency of virus core protein expression

Results

- Produced NP with a size around 110nm and had a good encapsulation efficiency percentage.

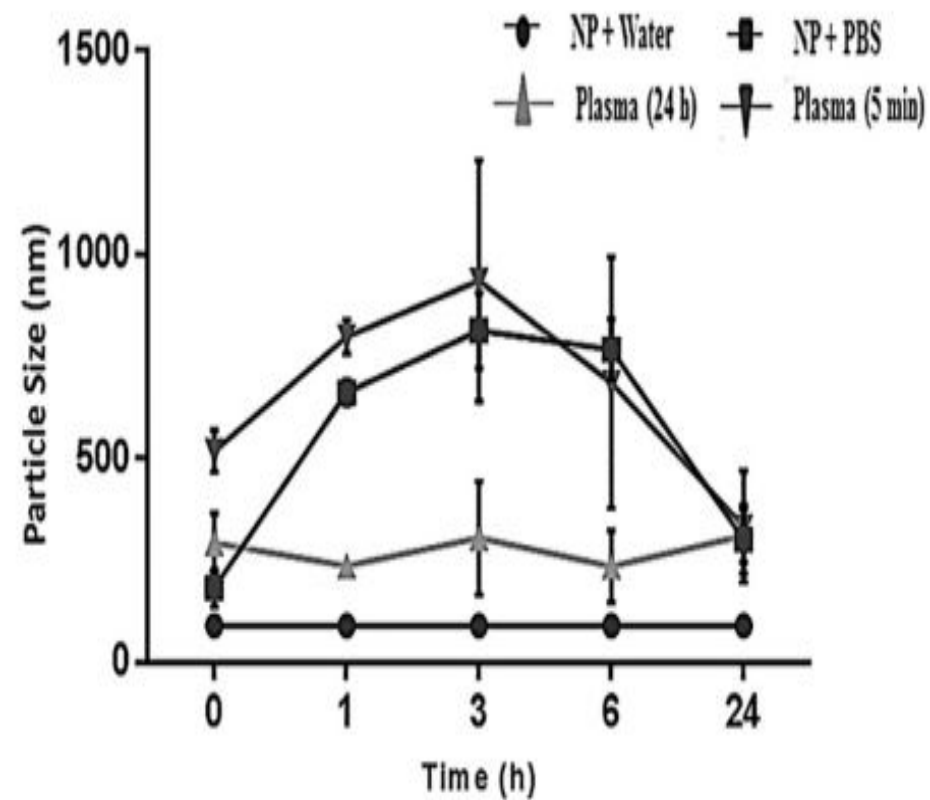
Physicochemical characterization of RBITC PGA NP.

Name	Particle Size (dnm $\pm$ SD) [*] [Pdl]	Zeta Potential (mv $\pm$ SD)	^{**} % Load % (w/w) $\pm$ SD	^{***} Encapsulation Efficiency % $\pm$ SD
RBITC PGA NP	110 $\pm$ 30[0.01]	-53.7 $\pm$ 13.34	0.54 $\pm$ 0.13	54 $\pm$ 13

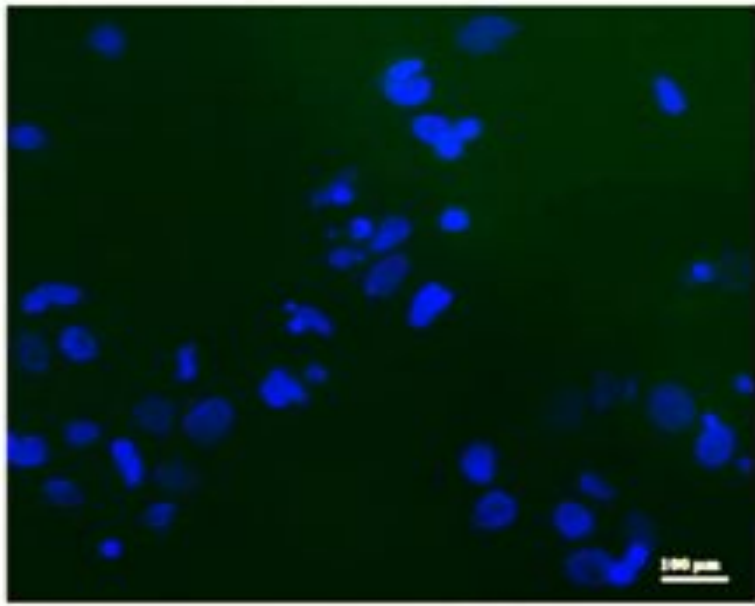
* Pdl = polydispersity index.

** % Load = [(Encapsulated dye weight/NP weight) * 100].

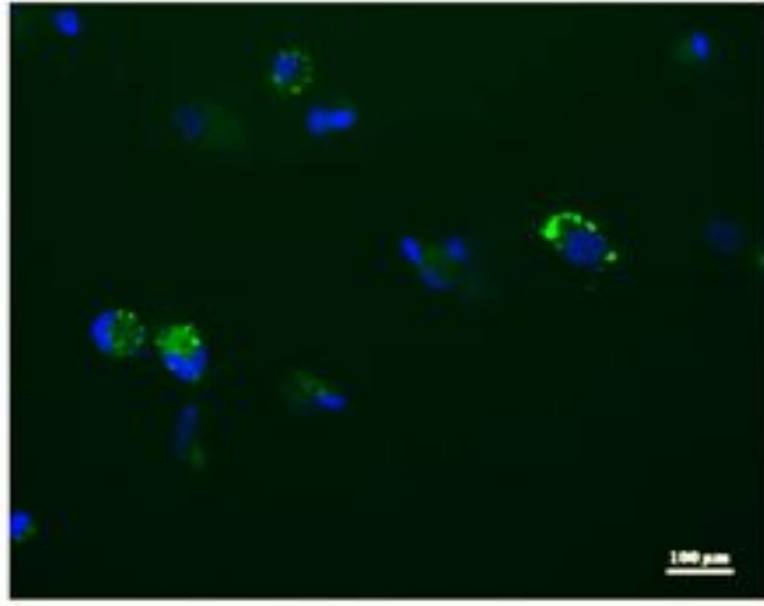
*** % Encapsulation = [(Encapsulated dye weight/Initial dye weight) * 100].



- 24h incubation w/ human plasma stabilized particles successfully in PBS
- Stability of NP in biological systems is an important consideration because particles in solutions with physiological salt concentrations and Ph values can form micrometer sizes coarse agglomerates.
- more efficient coating of NP by plasma protein came from the improvement of NP stabilization after 24 h with human plasma

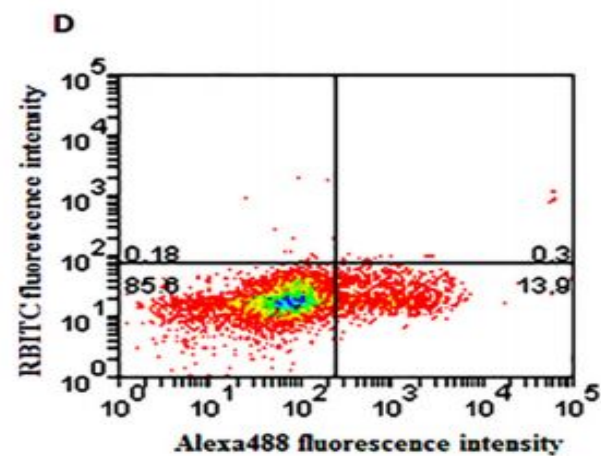
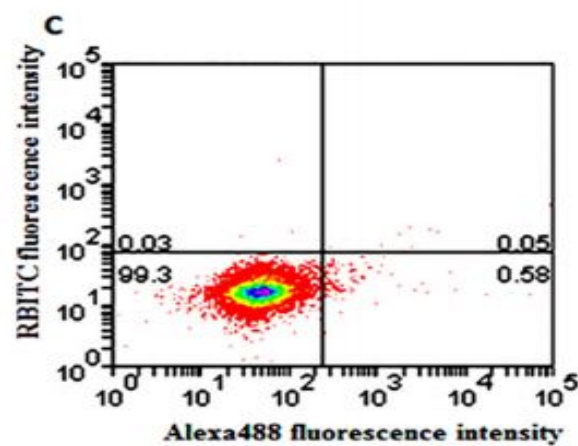
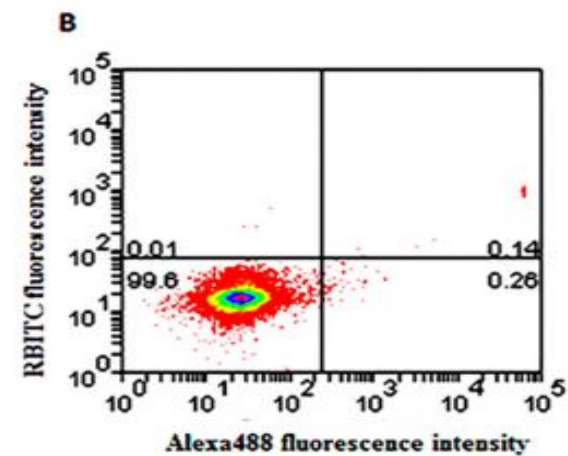
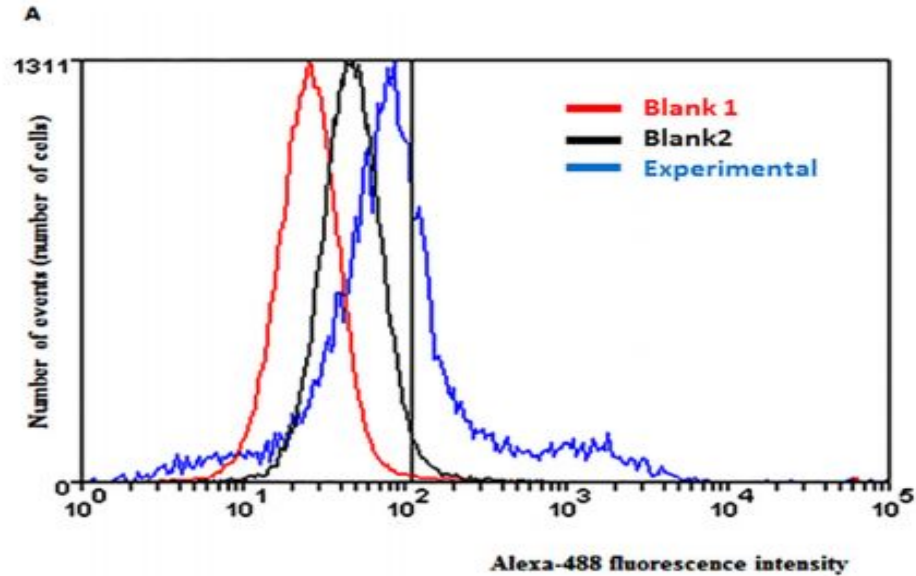


A

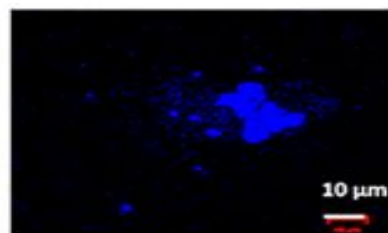


B

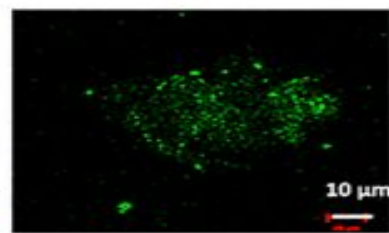
- Expression of virus core protein was detected 48h after transfection of Huh7.5 cells with J6/JFH1 and staining of the core protein using the primary antibody (C7-50) and secondary antibody (Alexa488)



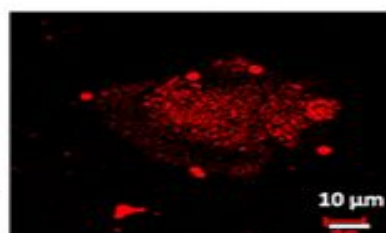
0.5h



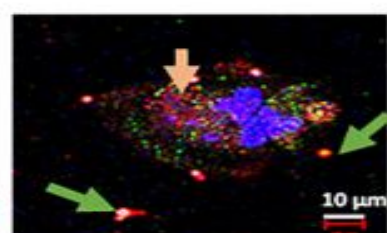
A



B

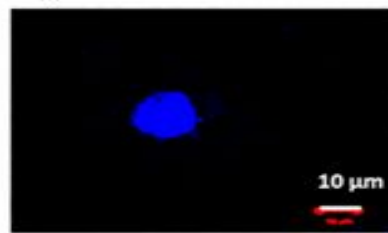


C

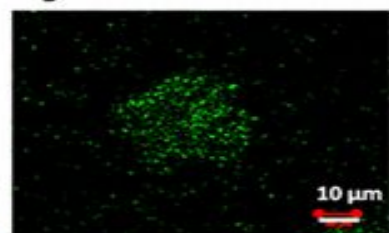


D

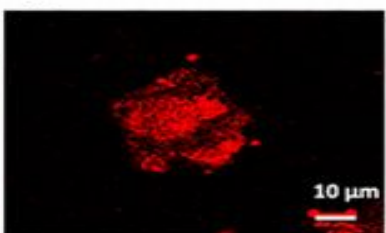
1h



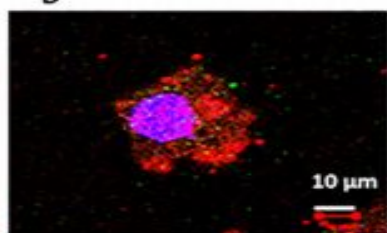
A



B

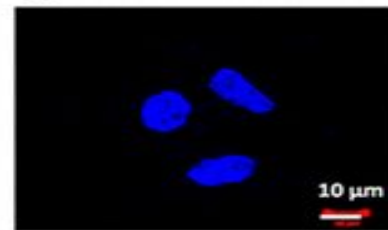


C

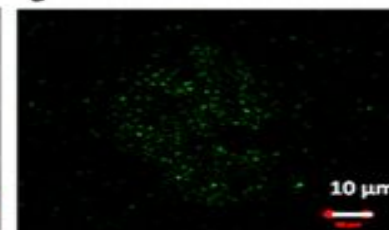


D

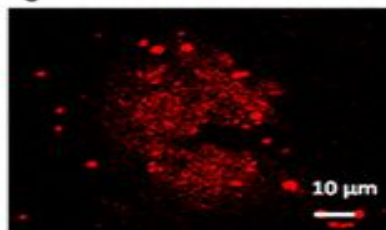
2h



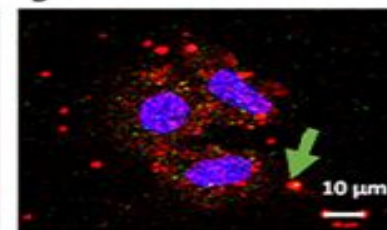
A



B

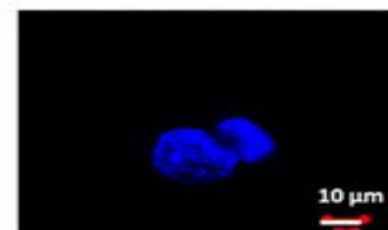


C

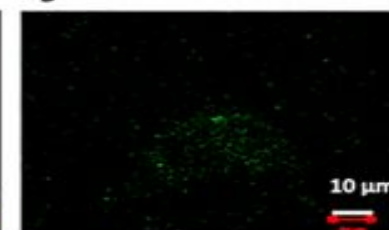


D

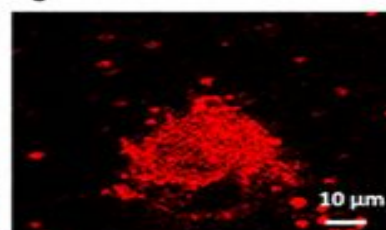
4h



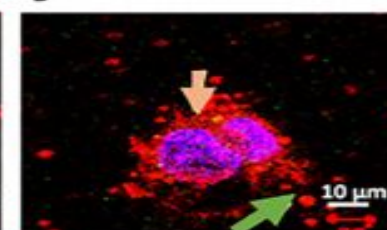
A



B



C



D

0.0 μm

2.5 μm

5.0 μm

7.6 μm

10.1 μm

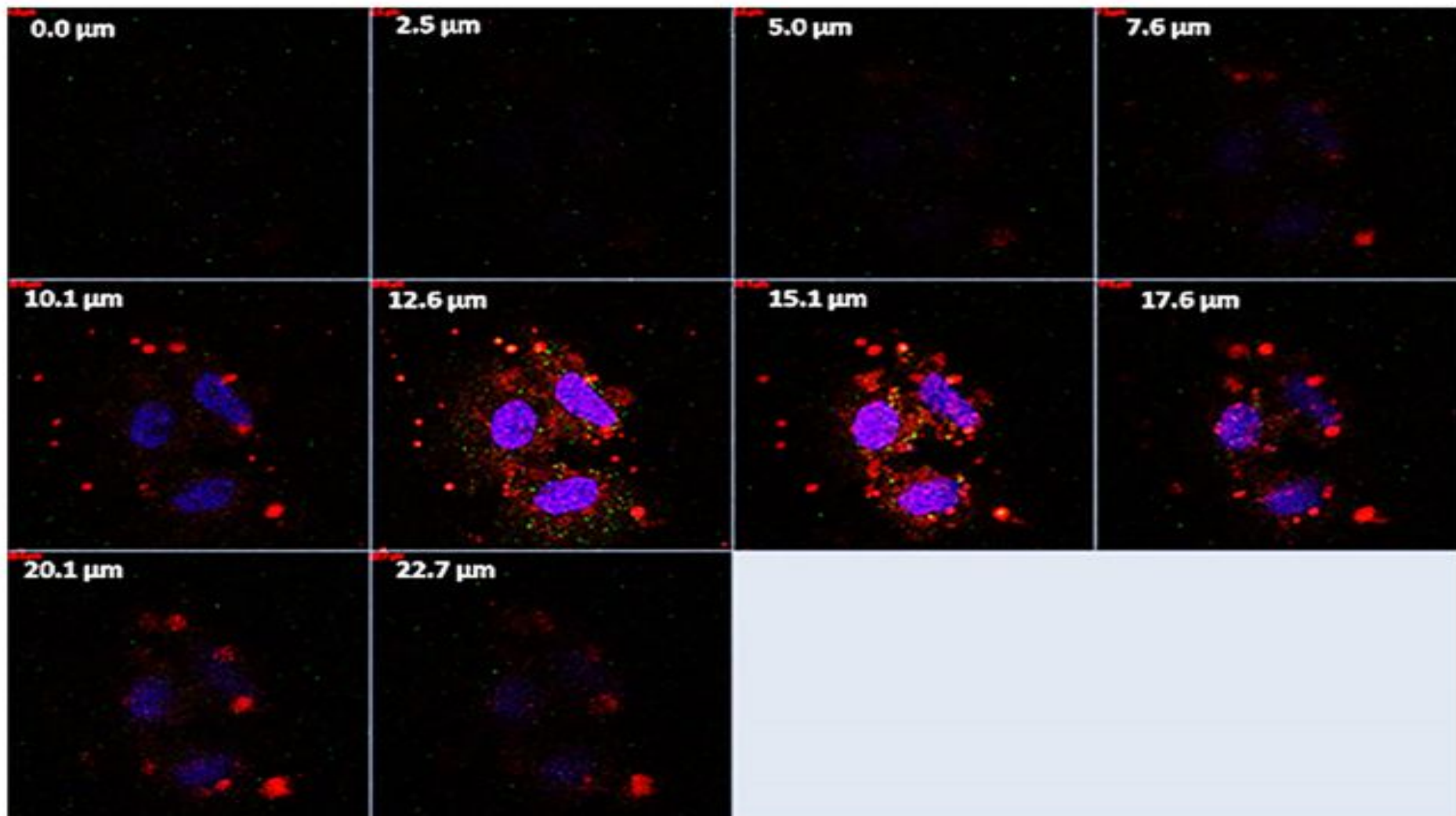
12.6 μm

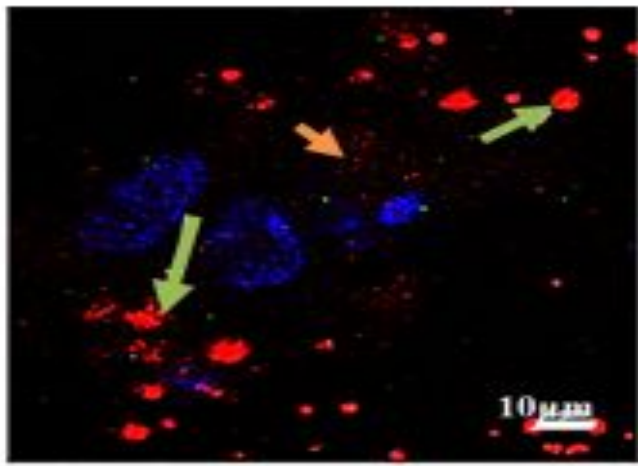
15.1 μm

17.6 μm

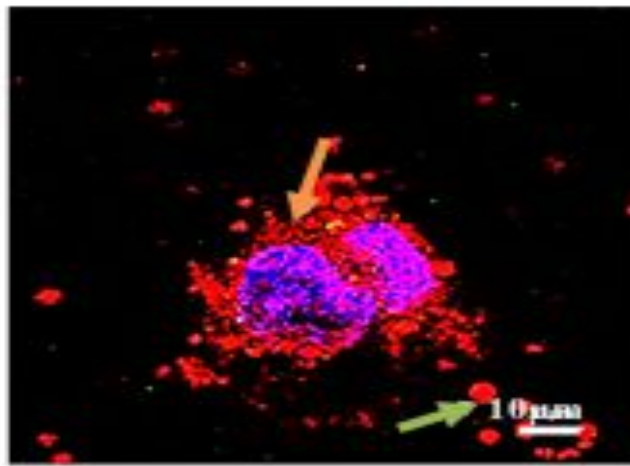
20.1 μm

22.7 μm





A



B

- A single plane confocal microscopy image of NP taken up by virus transfected cells compared to non transfected cells after incubation for 4 hours

Time (h)

Blank/
Control

0

0.5

1

2

4

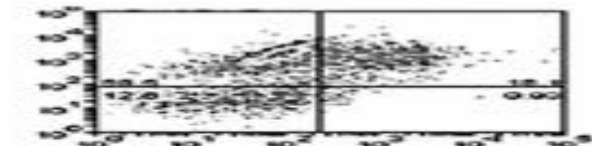
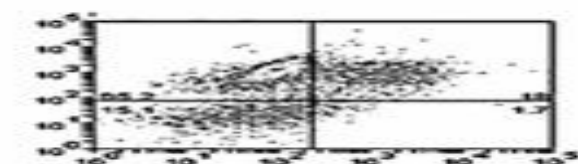
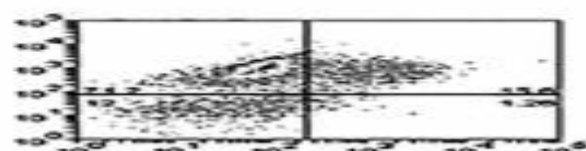
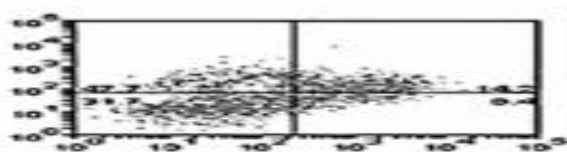
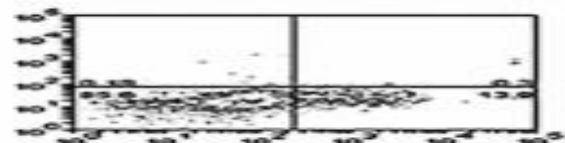
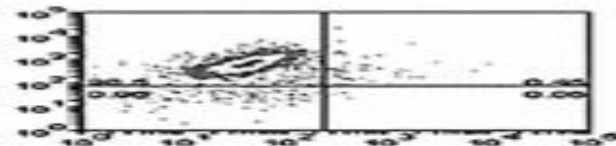
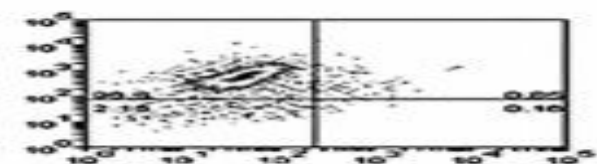
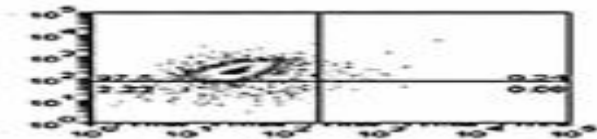
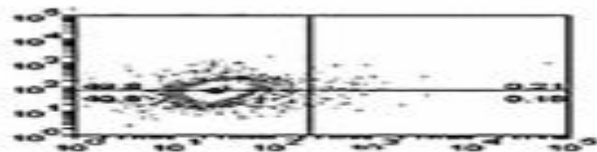


Table 2

% Gated cells for virus transfected and non-transfected Huh7.5 cells.

Time	%Gated for virus transfected Cells				%Gated for virus non-transfected cells			
	LL	UL	LR	UR	LL	UL	LR	UR
Experimental	99.34	0.03	0.58	0.05	-	-	-	-
0	31.38	51.64	4.30	12.67	27.14	72.53	0.11	0.22
0.5	13.74	66.77	1.65	17.84	2.46	97.17	0.09	0.29
1	14.15	67.82	1.60	16.42	1.47	98.11	0.09	0.32
2	13.03	67.32	1.43	18.22	1.46	97.91	0.13	0.51
3	11.10	69.18	0.82	18.90	1.09	98.46	0.07	0.38
4	10.78	68.60	0.91	19.70	1.04	98.50	0.07	0.39

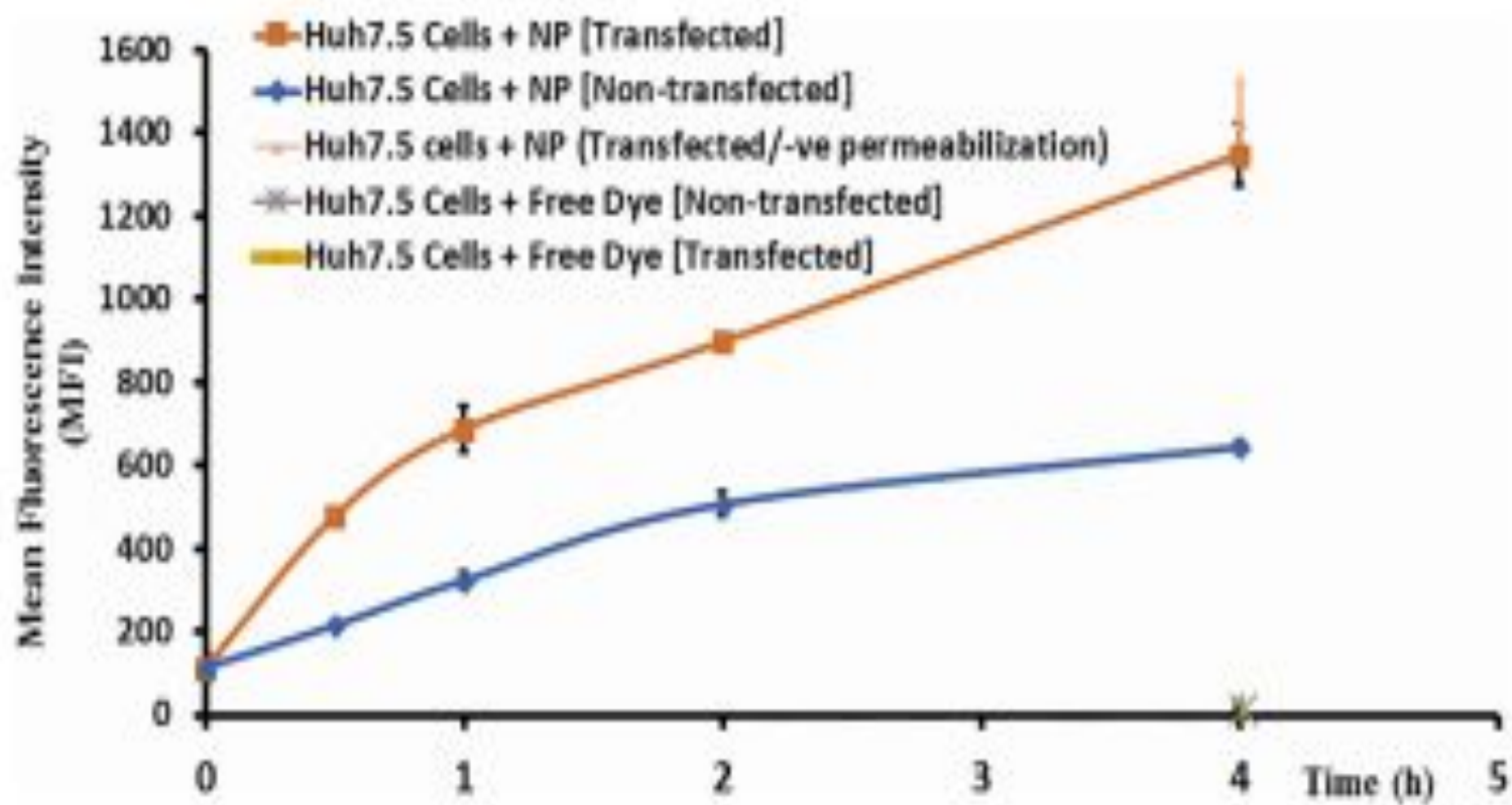
Experimental: Cells electroporated in presence of HCV RNA and treated with 1ry and 2ry antibodies.

LL: Lower Left quadrant; low virus transfection, low NP uptake.

UL: Upper Left quadrant; low virus transfection, raised NP uptake.

LR: Lower Right quadrant; raised virus transfection, low NP uptake.

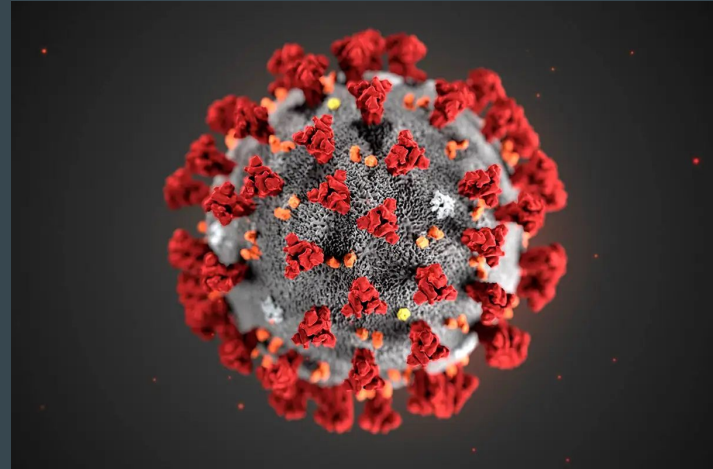
UR: Upper Right quadrant; raised virus transfection, raised NP uptake.



My proposal

Hypothesis:

- Nano delivery can be a good way to enhance delivery of antiviral agents to the throat/lungs while reducing side effects and off target accumulation in the case of virus infections such as COVID-19



Rationale for proposed study

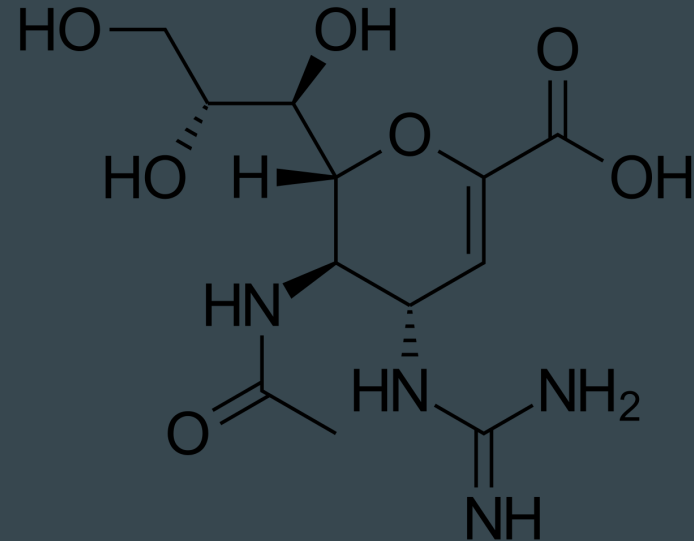
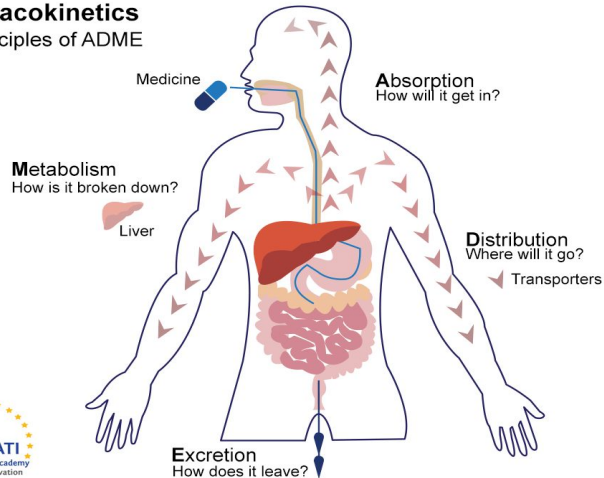
- Sung, Jean C, et al. "Nanoparticles for Drug Delivery to the Lungs." *Trends in Biotechnology*, U.S. National Library of Medicine, Dec. 2007, www.ncbi.nlm.nih.gov/pubmed/17997181.
- Cojocaru, Florina-Daniela, et al. "Nanomaterials Designed for Antiviral Drug Delivery Transport across Biological Barriers." *Pharmaceutics*, MDPI, 18 Feb. 2020, www.ncbi.nlm.nih.gov/pmc/articles/PMC7076512/.
- Boivin, et al. "Rapid Antiviral Effect of Inhaled Zanamivir in the Treatment of Naturally Occurring Influenza in Otherwise Healthy Adults." *OUP Academic*, Oxford University Press, 1 Apr. 2000, academic.oup.com/jid/article/181/4/1471/864974.

Aims

- Use Zanamivir to make nanoparticles into the form of micron-scale dry powder inhalations
- Use pharmacokinetic processes to ensure the medication gets delivered straight into the lungs

Pharmacokinetics

The principles of ADME



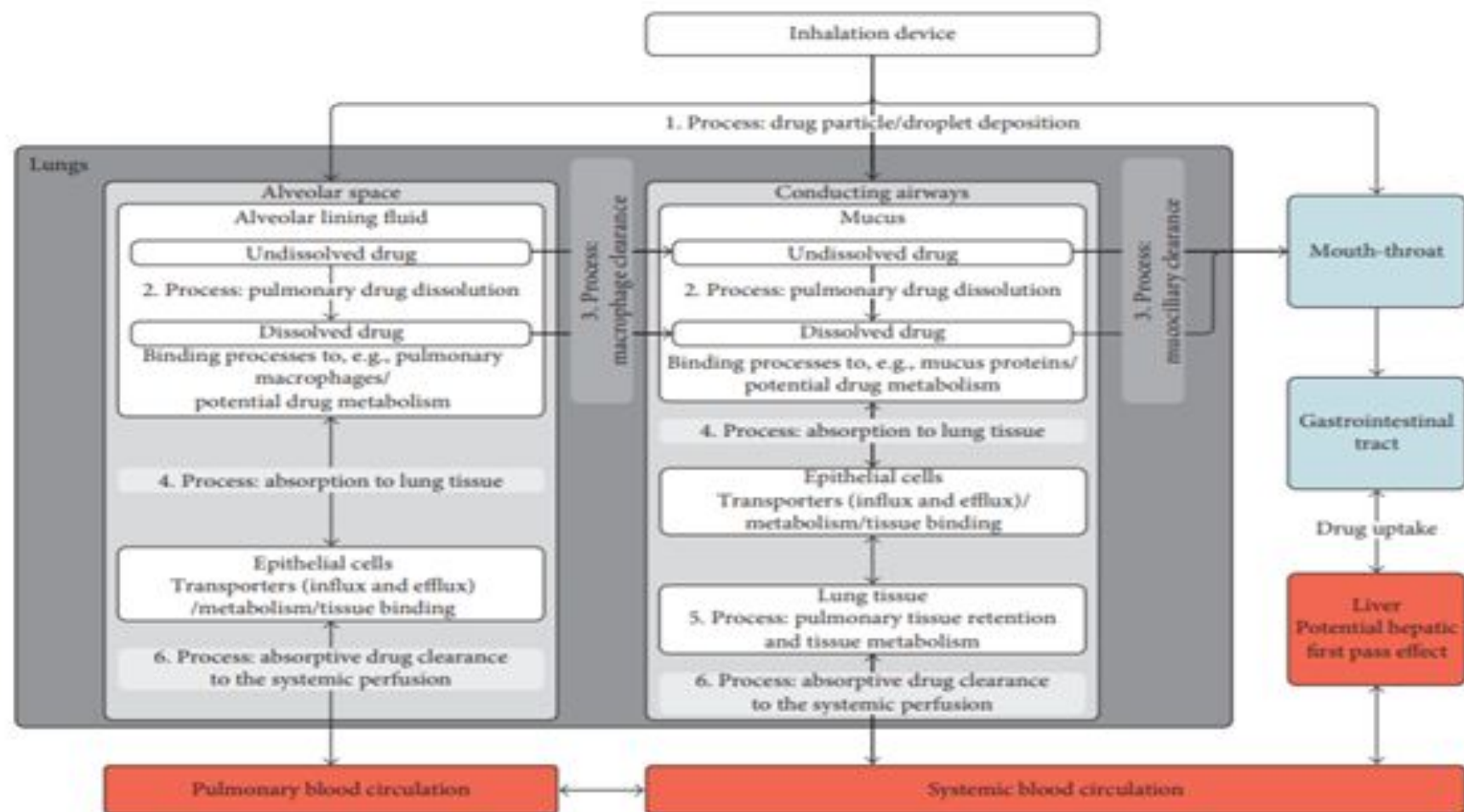


FIGURE 1: Summary of the lung-specific PK processes for inhaled drugs. Overview of the pulmonary-specific kinetic processes (1–6). The direction of the arrows indicates the direction of each process. For example, drug dissolution is considered to be a unidirectional process.

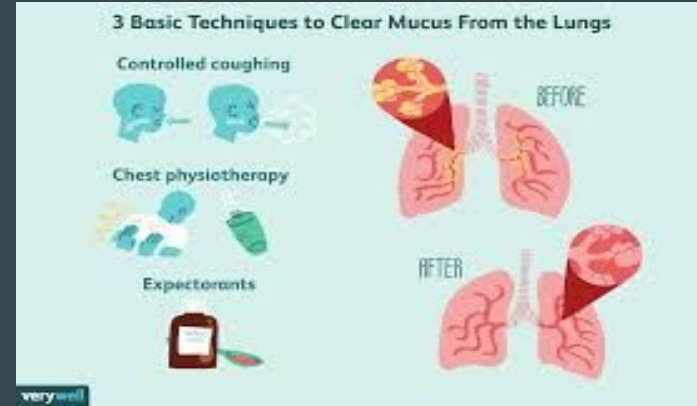
Challenges

- Formulation instability due to particle particle interactions
- Poor delivery efficiency due to exhalation of low-inertia nanoparticles
- Ensure that the nanoparticles get past the mucus tissue that's in the lungs and throat.



Methods

- Fast mucosal penetration or highly adhesive particles (slow turnover) to increase the drug's residence on the targeted mucosal
- Mucopenetration is the found to permeate the mucus layers by two potential mechanisms
 - Active strategy characterized by the interaction with the mucus and chemically shifting the features of the mucus or their own structures and passive strategy that uses hydrophilicity enhancers to penetrate the mucus.



**Thank
You**

